

Section: Particles and Waves

Topic: Interference

A student sets up a **double-slit interference** experiment using a laser and a screen.

(a)

State the effect observed on the screen

 A **pattern of bright and dark fringes** (interference pattern) is observed.
These are caused by **constructive and destructive interference** of the light from the two slits.

(b)

State two changes that could be made to increase the fringe separation

Fringe separation w is given by:

$$w = \frac{\lambda D}{d}$$

Where:

- λ : wavelength of light
- D : distance from slits to screen
- d : slit separation

To increase w , you can:

1. **Increase the wavelength** (e.g. use redder light)
2. **Increase the distance to the screen**

Alternatively:

3. **Decrease the slit separation**

 Any two of these will increase fringe spacing.

(c)

State why laser light is used in this experiment

 Laser light is **monochromatic** (single wavelength) and **coherent** (constant phase difference), making it ideal for producing a **clear, stable interference pattern**.

 Revision Tips

- The **interference pattern** depends on path difference and phase difference
- Coherent sources are essential for a clear pattern
- Fringe spacing increases with:
 - $\uparrow$ wavelength
 - $\uparrow$ screen distance
 - $\downarrow$ slit separation

- Use:

$$w = \frac{\lambda D}{d}$$

to analyse changes